Root [0]

Technical Sheet Only!
Please refer to the subsequent sheets for schematic

Power
File: power.kicad_sch

FPGA_Power
File: fpga_power.kicad_sch

FPGA_Config
File: fpga_config.kicad_sch

JTAG
File: jtag.kicad_sch

Serial
File: serial.kicad_sch

USB
File: usb.kicad_sch

Connectivity
File: connectivity.kicad_sch

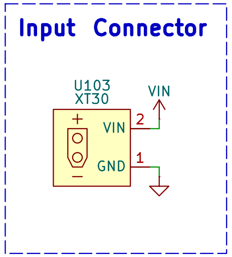
Input
File: voltage_input.kicad_sch

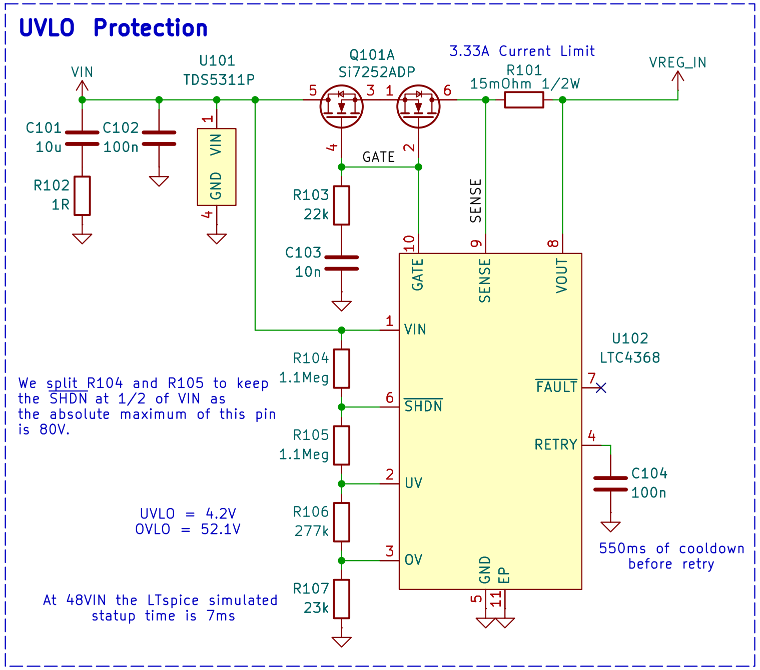
Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /
File: Controller_Board.kicad_sch
Title: Root

Size: A4	Date: 2026-03-10	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 0/9

Input [1]

Maybe connect the P8_RX J4 and H4 and use the Bank 35 at 3V3





In[2]:= $R3 = \frac{3*10^{-3}}{10*10^{-9}} * \left(\frac{4-0.5}{0.5}\right)$

Out[2]= 2.1×10^{6}

In[3]:= $R1 = \frac{\left(\frac{3*10^{-3}}{10*10^{-9}}\right) + 2.1*10^{6}}{52} * 0.5$

Out[3]= 23076.9

In[4]:= $R2 = \left(\frac{3*10^{-3}}{10*10^{-9}}\right) - 23076.9$

Out[4]= 276923.

In[14]:= $\text{NSolve}\left[2.2*10^{6} == \frac{3*10^{-3}}{10*10^{-9}} * \left(\frac{UV-0.5}{0.5}\right), UV\right]$

Out[14]= $\{\{UV \to 4.16667\}\}$

In[15]:= $\text{NSolve}\left[24000 == \frac{\left(\frac{3*10^{-3}}{10*10^{-9}}\right) + 2.2*10^{6}}{OV} * 0.5, OV\right]$

Out[15]= $\{\{OV \to 52.0833\}\}$

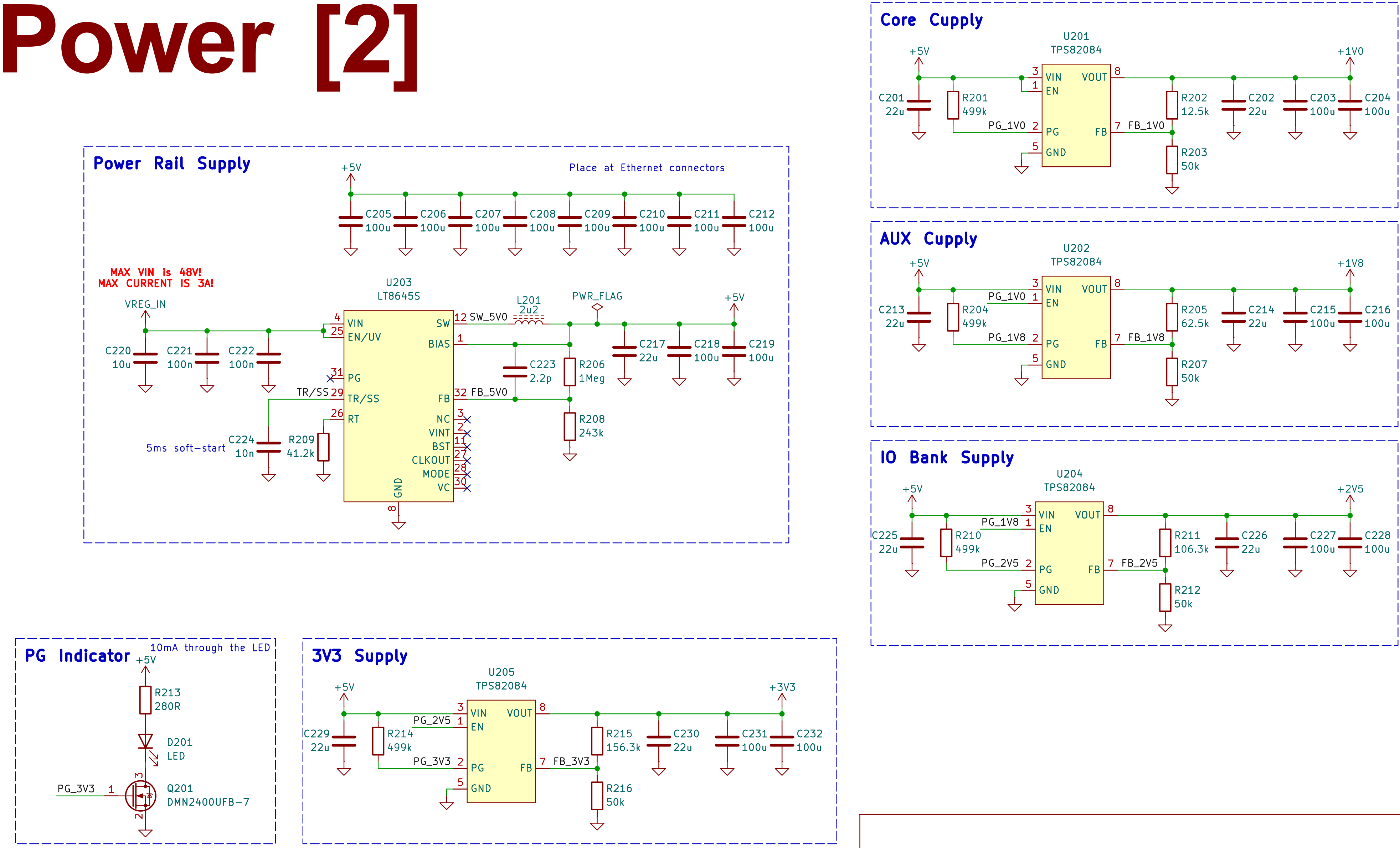
UV/OV Calculation

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /Input/
File: voltage_input.kicad_sch

Title: Input

Size: A4	Date: 2026-03-27	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 1/9

Power [2]

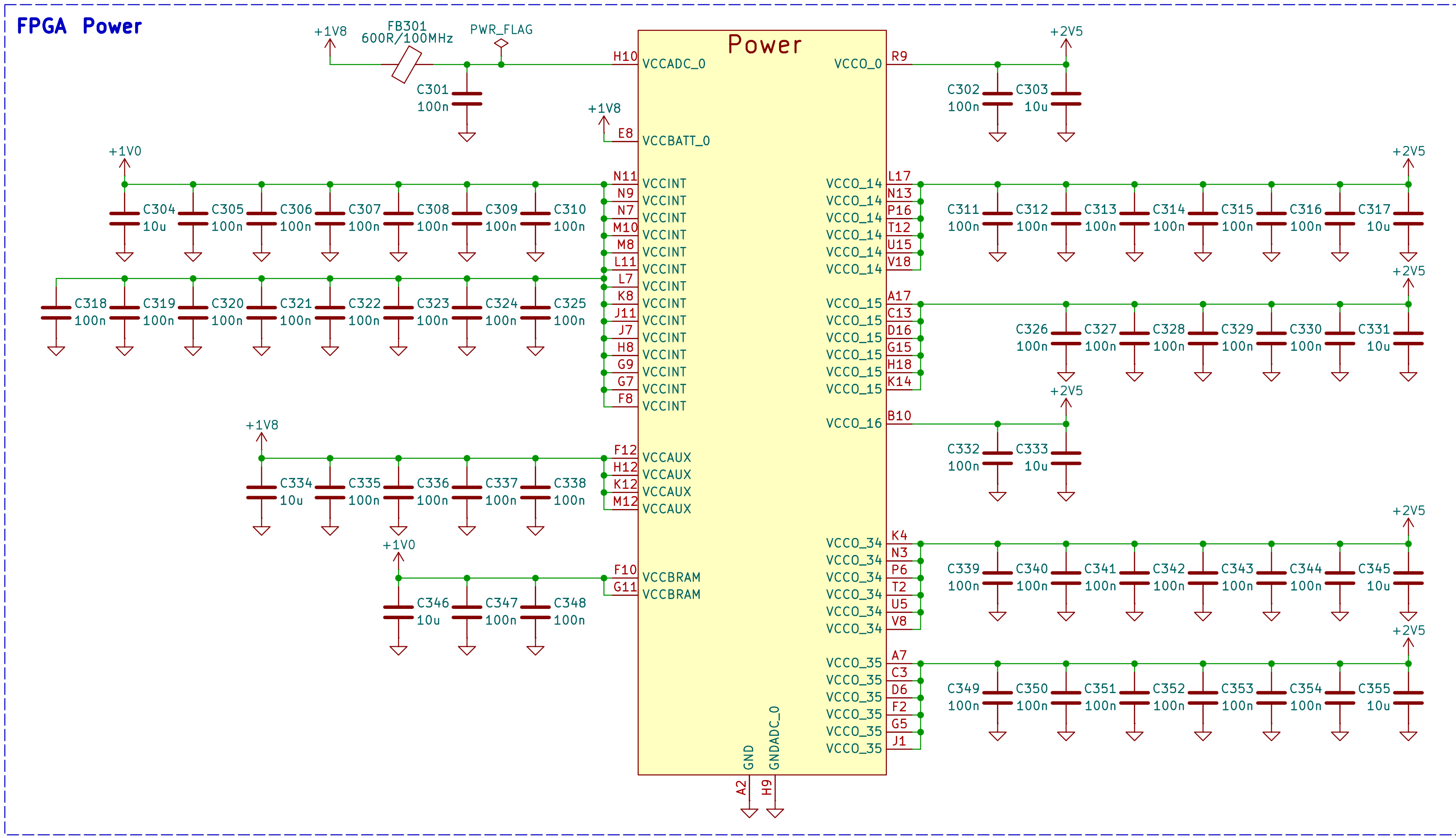


Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /Power/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-03-07	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 2/9

FPGA Power [3]



Power Supply

Source	Voltage	Total (A)
V_{CCINT}	1.000	0.510
V_{CCBRAM}	1.000	0.001
V_{CCAUX}	1.800	0.112
V_{CCAUX_IO}		
V_{CCO} 3.3V	3.300	
V_{CCO} 2.5V	2.500	0.861
V_{CCO} 1.8V	1.800	
V_{CCO} 1.5V	1.500	
V_{CCO} 1.35V	1.350	
V_{CCO} 1.2V	1.200	
$MGTV_{CCAUX}$	1.800	
$MGTAV_{CC}$	1.000	
$MGTAV_{TT}$	1.200	
-		
-		
-		
-		
-		
-		
V_{CCADC}	1.800	0.020

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /FPGA_Power/
File: fpga_power.kicad_sch
Title: FPGA Power
Size: A4 Date: 2026-03-10 **Rev: 1.0**
KiCad E.D.A. 10.0.0 Id: 3/9

FPGA Config [4]

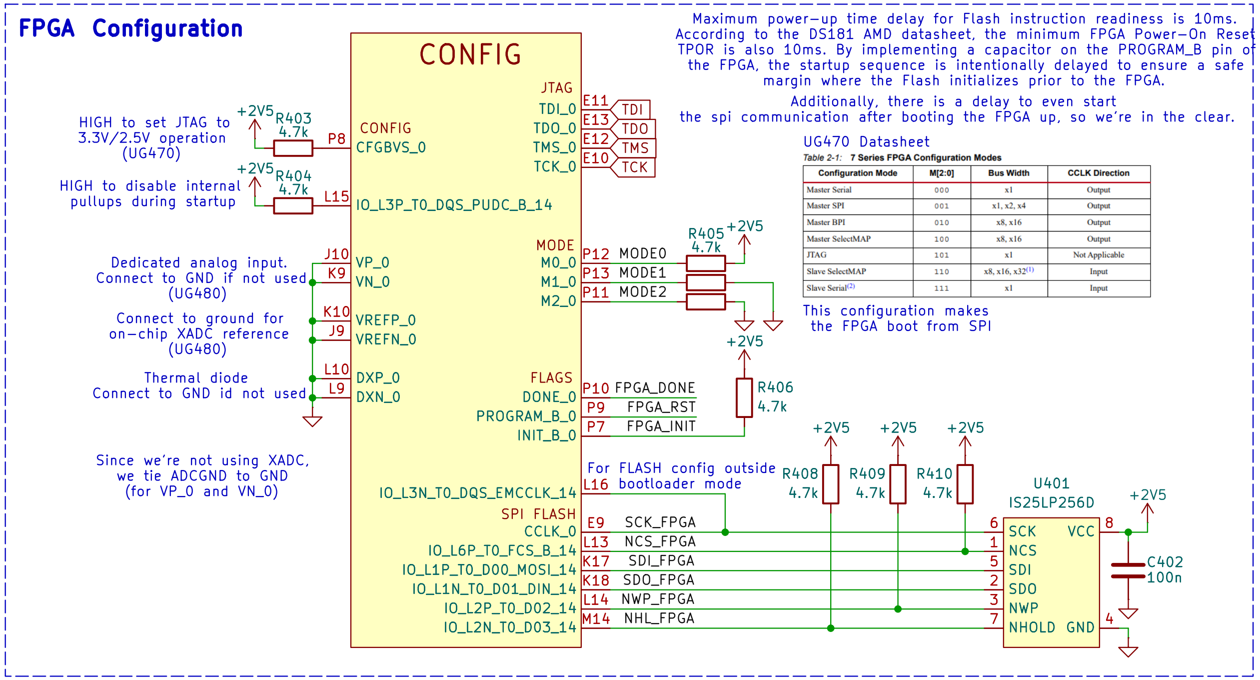
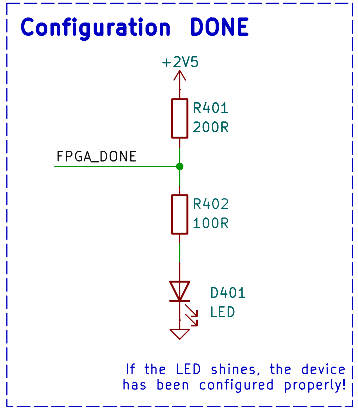
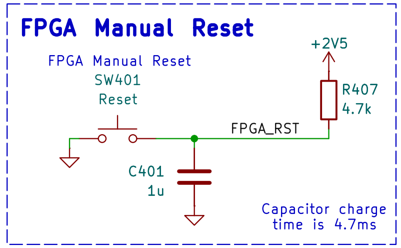
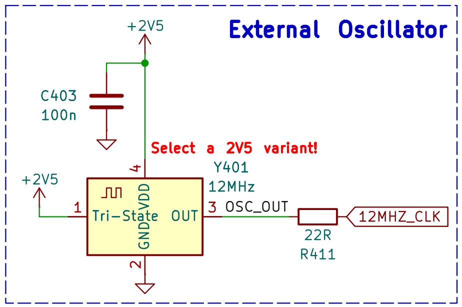


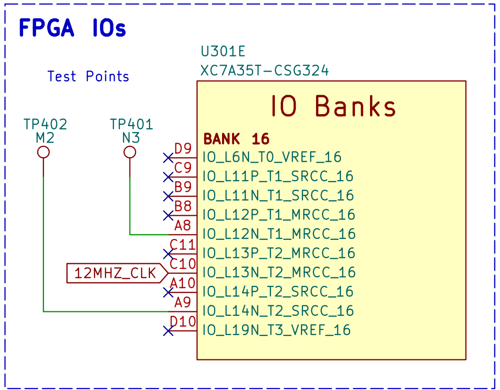
Table 2-1: **7 Series FPGA Configuration Modes**

Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master Serial	000	x1	Output
Master SPI	001	x1, x2, x4	Output
Master BPI	010	x8, x16	Output
Master SelectMAP	100	x8, x16	Output
JTAG	101	x1	Not Applicable
Slave SelectMAP	110	x8, x16, x32[1]	Input
Slave Serial[2]	111	x1	Input









Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague

Sheet: /FPGA_Config/
File: fpga_config.kicad_sch

Title: FPGA Config		
Size: A4	Date: 2026-03-10	**Rev: 1.0**
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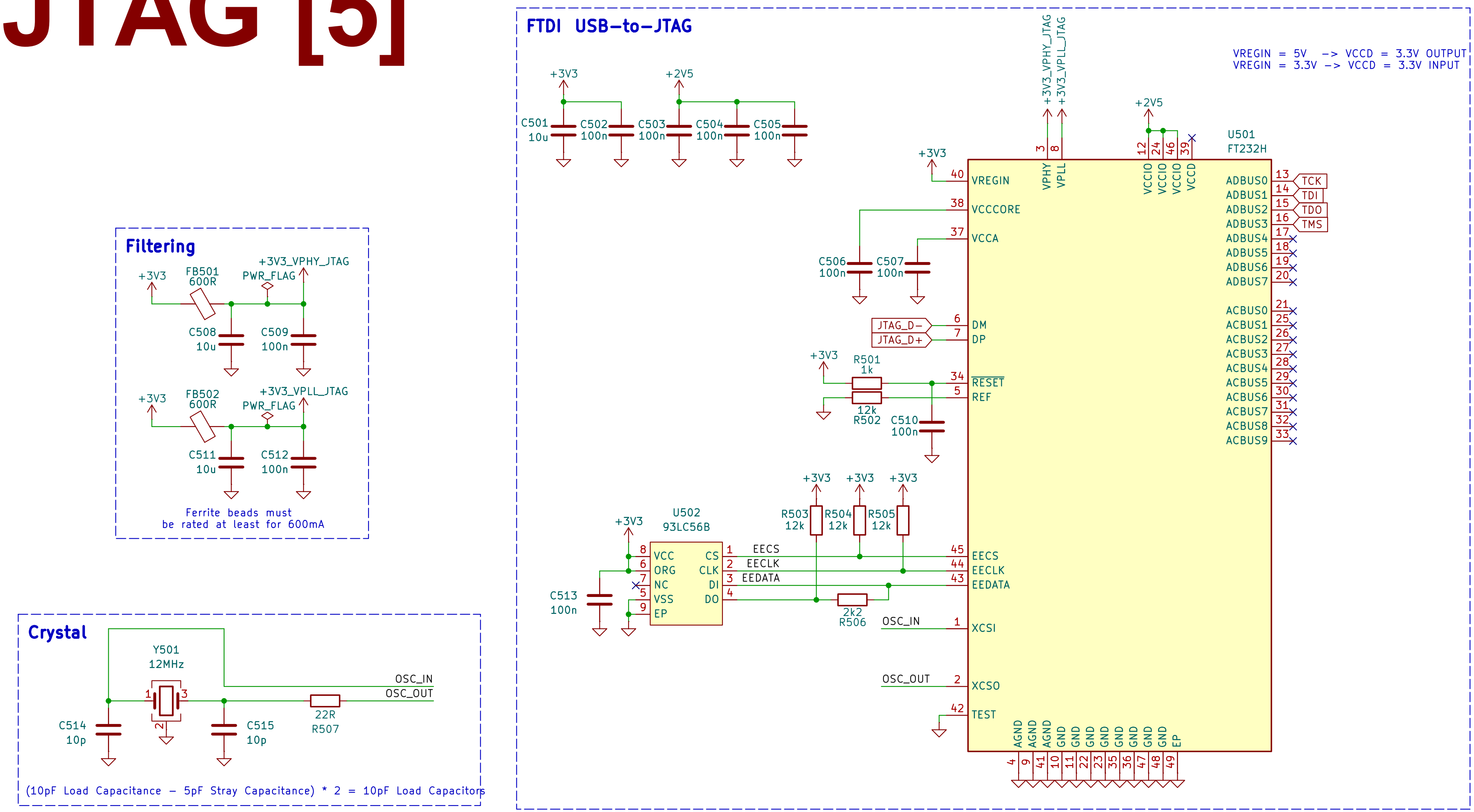
JTAG [5]



Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /JTAG/
File: jtag.kicad_sch

Title: JTAG

Size: A4 | Date: 2026-03-11 | **Rev: 1.0**
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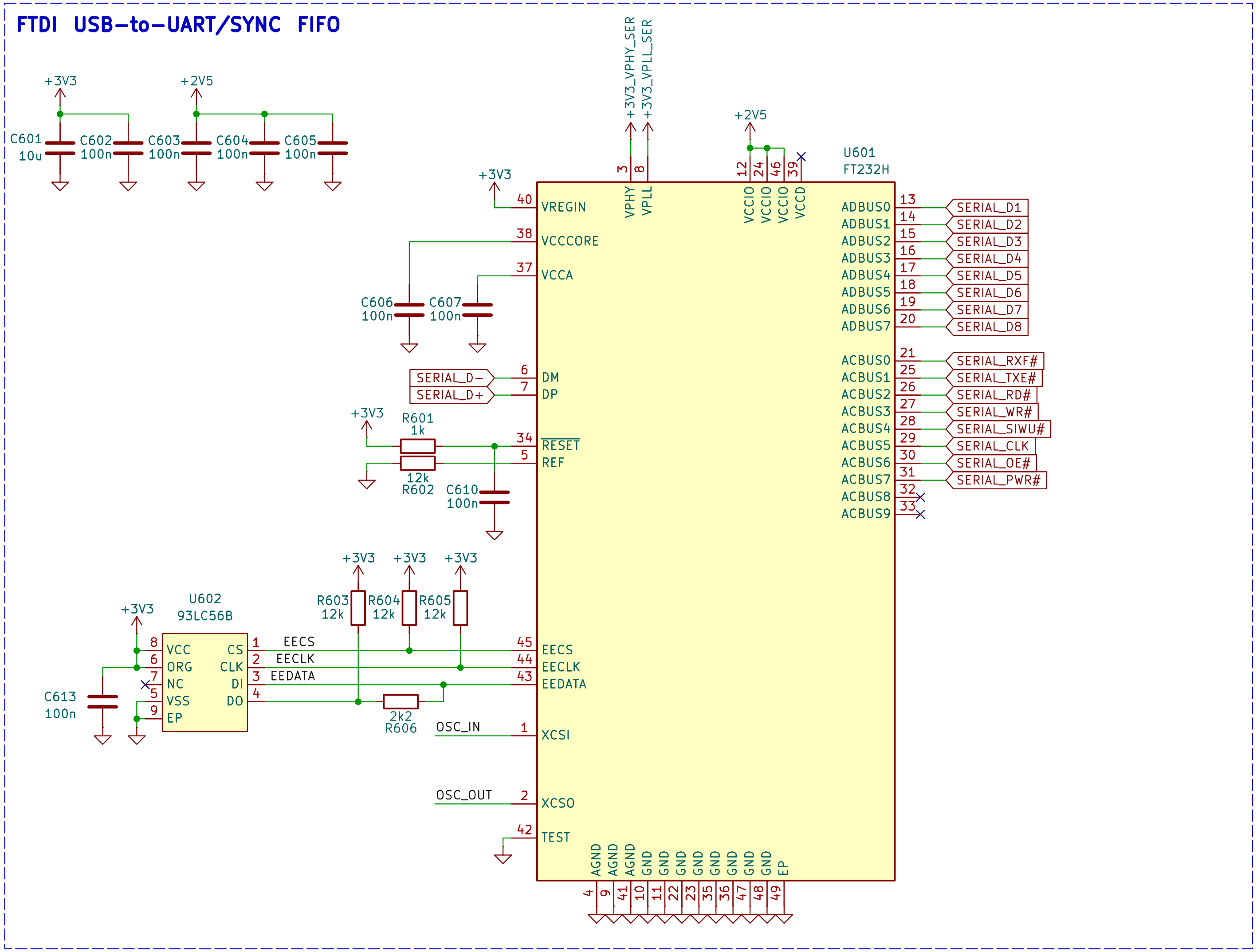
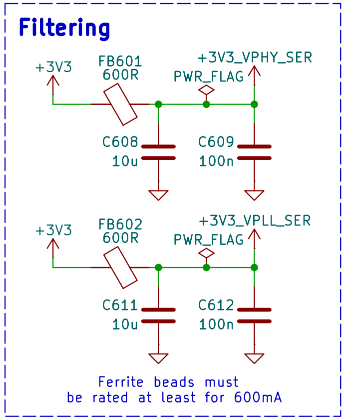
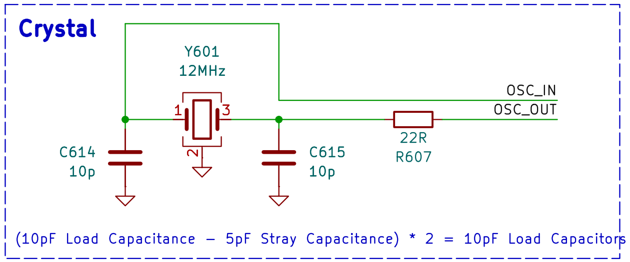
Serial [6]







Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /Serial/
File: serial.kicad_sch

Title: Serial

Size: A4 | Date: 2026-03-11 | **Rev: 1.0**
KiCad E.D.A. 10.0.0 | Id: 6/9

USB [7]

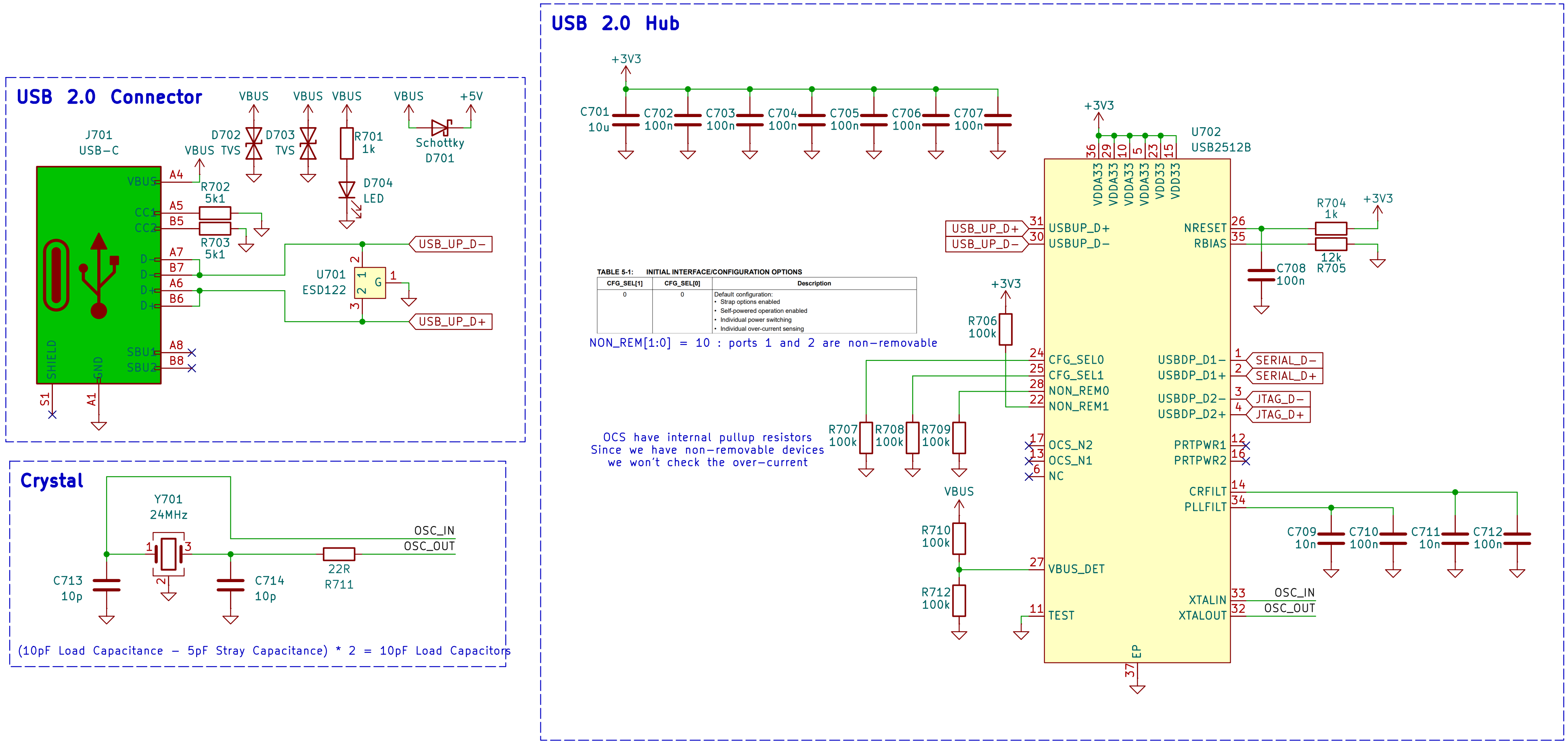


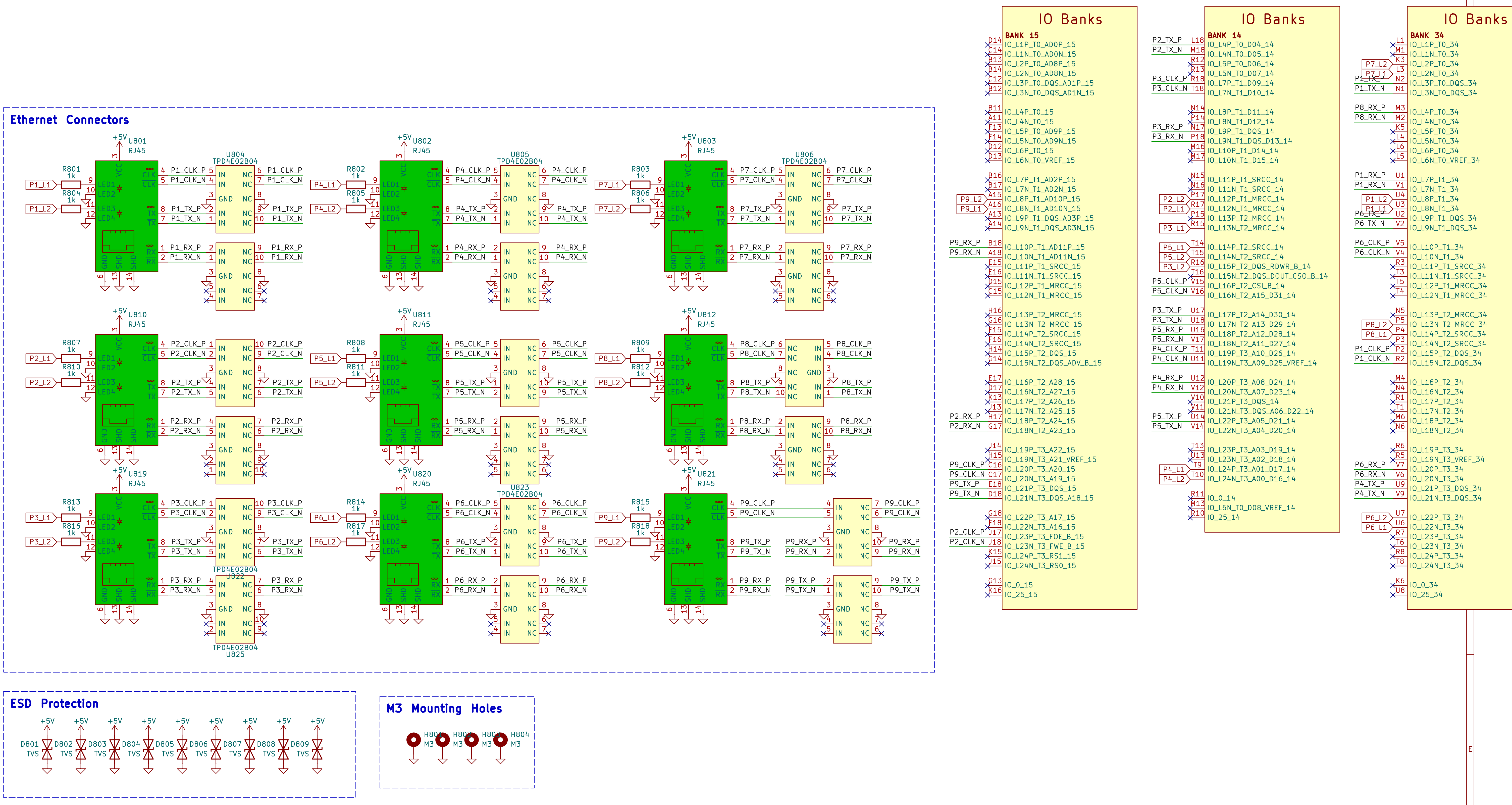
TABLE 5-1: INITIAL INTERFACE/CONFIGURATION OPTIONS

CFG_SEL[1]	CFG_SEL[0]	Description
0	0	Default configuration: • Strap options enabled • Self-powered operation enabled • Individual power switching • Individual over-current sensing

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /USB/
File: usb.kicad_sch
Title:
Size: A4 Date:
KiCad E.D.A. 10.0.0
Rev: 1.0
Id: 7/9

Connectivity [8]

Please exuse this mess, I'll try to make this pretty later



Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague
Sheet: /Connectivity/
File: connectivity.kicad_sch

Title: Connectivity

Size: A3	Date: 2026-03-12	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 8/9